18.404/6.840 Lecture 2

Last time: (Sipser §1.1)

- Finite automata, regular languages
- Regular operations $\cup, \circ, *$
- Regular expressions
- Closure under $\cup$

Today: (Sipser §1.2 – §1.3)

- Nondeterminism
- Closure under $\circ$ and $*$
- Regular expressions → finite automata

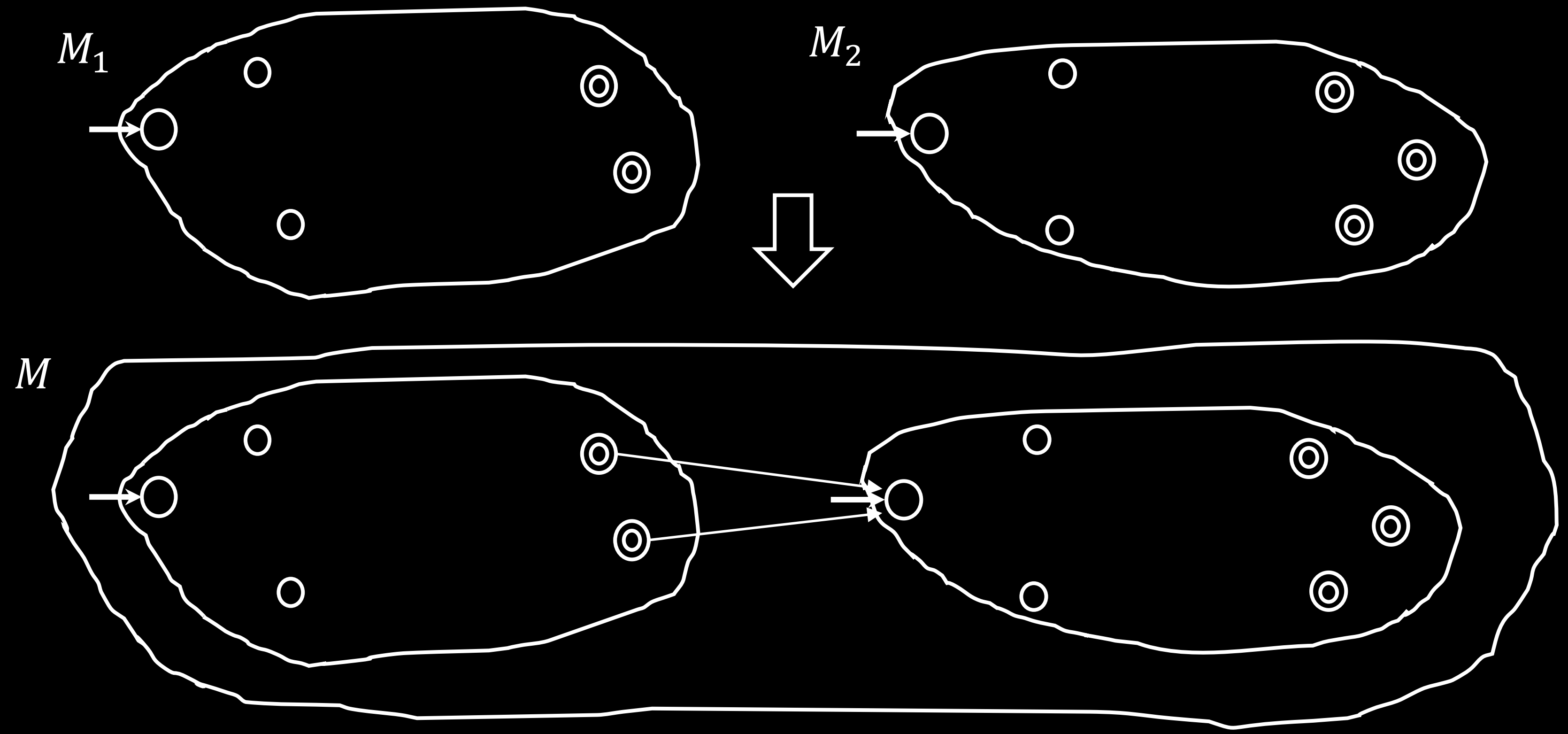
Goal: Show finite automata equivalent to regular expressions

Closure Properties for Regular Languages

Theorem: If A_1, A_2 are regular languages, so is $A_1 A_2$ (closure under $\circ$)

Recall proof attempt: Let $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1
$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize A_2

Construct $M = (Q, \Sigma, \delta, q_0, F)$ recognizing $A_1 A_2$

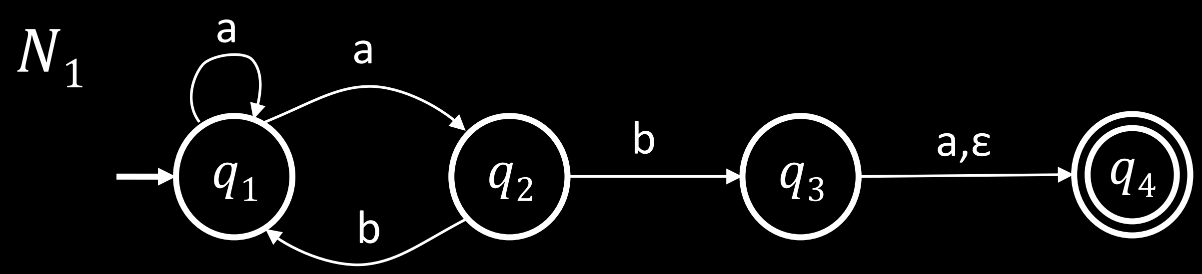


M should accept input w
if $w = xy$ where
M_1 accepts x and M_2 accepts y.

Doesn't work: Where to split w?

Hold off. Need new concept. 

Nondeterministic Finite Automata



New features of nondeterminism:

- multiple paths possible (0, 1 or many at each step)
- ε-transition is a "free" move without reading input
- Accept input if <u>some</u> path leads to accept

Example inputs:

- ab
- aa
- aba
- abb

Check-in 2.1

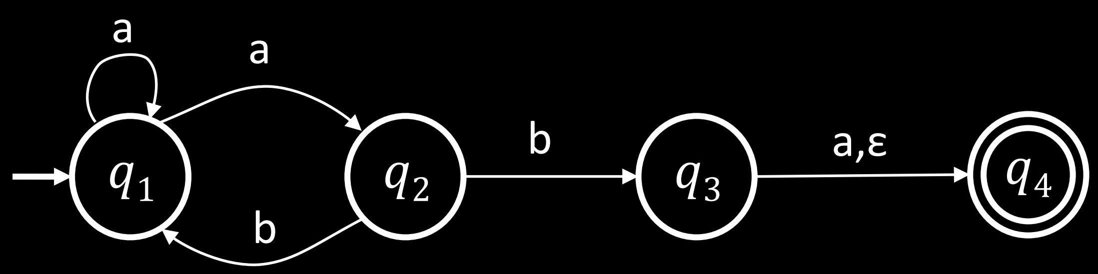
What does N_1 do on input aab ?

(a) Accept
(b) Reject
(c) Both Accept and Reject

Check-in 2.1

NFA – Formal Definition

N_1



Defn: A <u>nondeterministic finite automaton (NFA)</u>

N is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$

states
alphabet
transition function
start state
accept states

- all same as before except δ
- $\delta: Q \times \underbrace{\Sigma_\varepsilon}_{\Sigma \cup \{\varepsilon\}} \to \underbrace{\boldsymbol{\mathcal{P}}(Q)}_{\text{power set}} = \{R | R \subseteq Q\}$
- In the N_1 example: $\delta(q_1, \mathrm{a}) = \{q_1, q_2\}$
  $\delta(q_1, \mathrm{b}) = \emptyset$

Ways to think about nondeterminism:

<u>Computational:</u> Fork new parallel thread and accept if any thread leads to an accept state.

<u>Mathematical:</u> Tree with branches.
Accept if any branch leads to an accept state.

<u>Magical:</u> Guess at each nondeterministic step which way to go. Machine always makes the right guess that leads to accepting, if possible.

Converting NFAs to DFAs

Theorem: If an NFA recognizes A then A is regular

Proof: Let NFA $M = (Q, \Sigma, \delta, q_0, F)$ recognize A

Construct DFA $M' = (Q', \Sigma, \delta', q_0', F')$ recognizing A

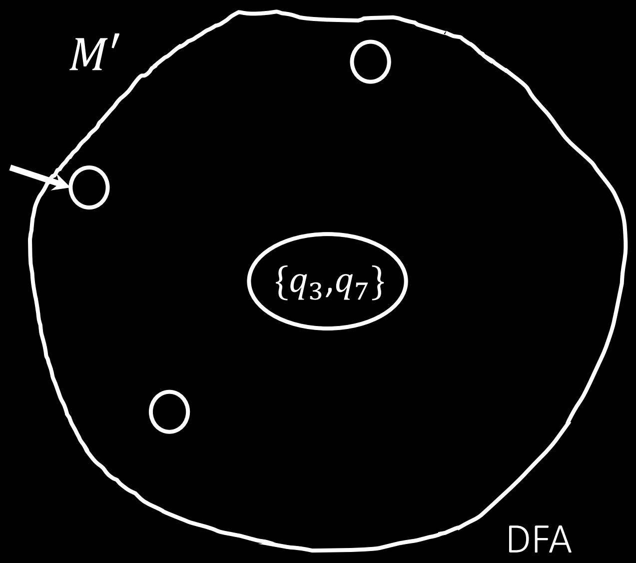
(Ignore the ε-transitions, can easily modify to handle them)

IDEA: DFA M' keeps track of the subset of possible states in NFA M.

Check-in 2.2

If M has n states, how many states does M' have by this construction?

(a) $2n$
(b) n^2
(c) 2^n



Construction of M':

$Q' = \mathcal{P}(Q)$

$\delta'(\underbrace{R}_{R \in Q'}, a) =$

$q_0' = \{q_0\}$

$F' = \{R \in Q' \mid R \text{ intersects } F\}$

Check-in 2.2

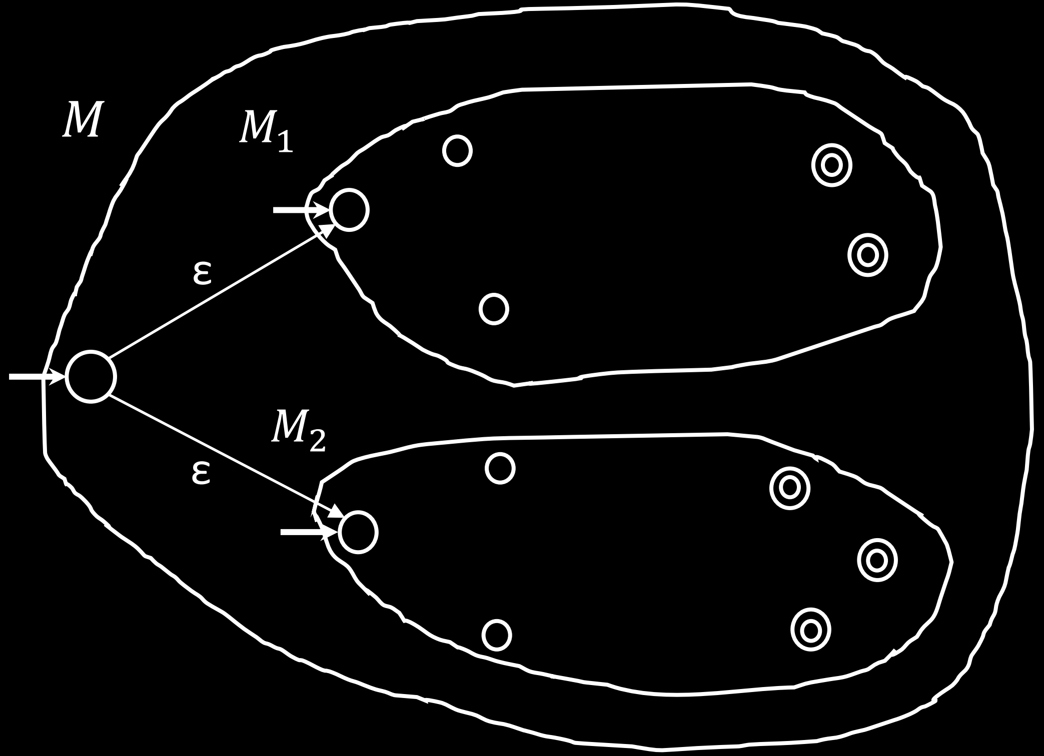
Return to Closure Properties

Recall Theorem: If A_1, A_2 are regular languages, so is $A_1 \cup A_2$

(The class of regular languages is closed under union)

New Proof (sketch): Given DFAs M_1 and M_2 recognizing A_1 and A_2

Construct NFA M recognizing $A_1 \cup A_2$

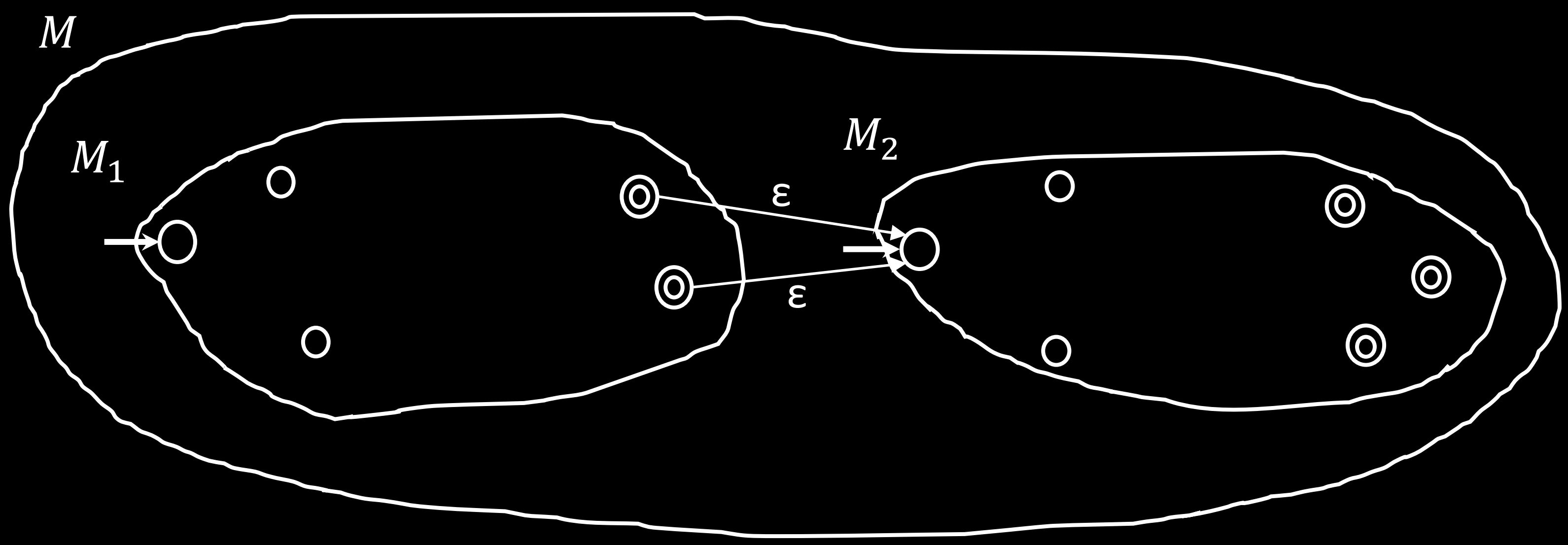


<u>Nondeterminism</u>
parallelism
vs
guessing

Closure under ∘ (concatenation)

Theorem: If A_1, A_2 are regular languages, so is $A_1 A_2$

Proof sketch: Given DFAs M_1 and M_1 recognizing A_1 and A_2
Construct NFA M recognizing $A_1 A_2$



M should accept input w
if $w = xy$ where
M_1 accepts x and M_2 accepts y.

$w =$ x | y

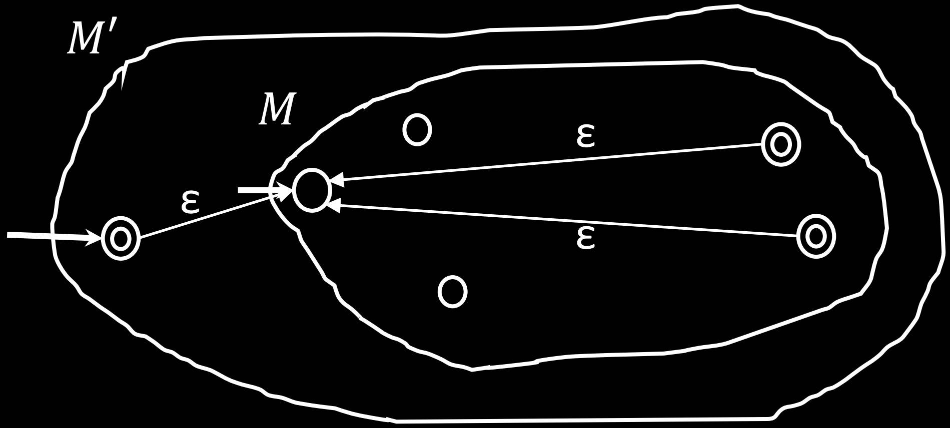
Nondeterministic M' has <u>the option</u>
to jump to M_2 when M_1 accepts.

Closure under $*$ (star)

Theorem: If A is a regular language, so is A^*

Proof sketch: Given DFA M recognizing A

Construct NFA M' recognizing A^*



Make sure M' accepts ε

Check-in 2.3

If M has n states, how many states does M' have by this construction?

(a) n

(b) $n + 1$

(c) $2n$

Check-in 2.3

Regular Expressions → NFA

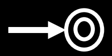
Theorem: If R is a regular expr and $A = L(R)$ then A is regular

Proof: Convert R to equivalent NFA M:

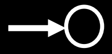
If R is <u>atomic:</u> Equivalent M is:

$R = a$ for $a \in \Sigma$

$R = \varepsilon$



$R = \emptyset$



If R is <u>composite:</u>

$R = R_1 \cup R_2$

$R = R_1 \circ R_2$

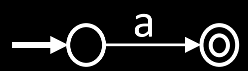
$R = R_1^*$

} Use closure constructions

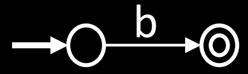
Example:

Convert $(\mathrm{a} \cup \mathrm{ab})^*$ to equivalent NFA

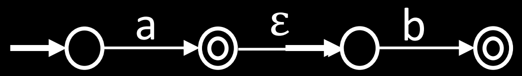
a:



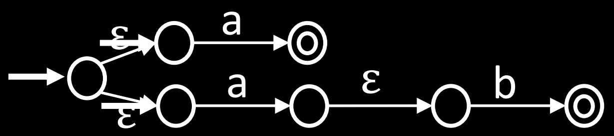
b:



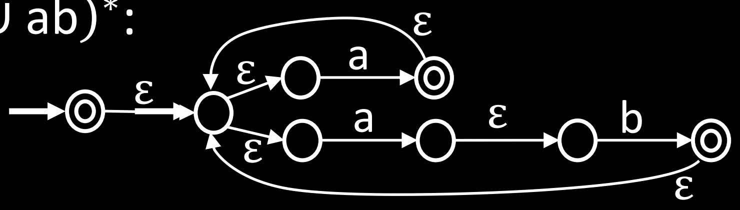
ab:



a ∪ ab:



$(\mathrm{a} \cup \mathrm{ab})^*$:



Quick review of today

1. Nondeterministic finite automata (NFA)
2. Proved: NFA and DFA are equivalent in power
3. Proved: Class of regular languages is closed under $\circ, *$
4. Conversion of regular expressions to NFA

Check-in 2.4

Recitations start tomorrow online (same link as for lectures).
They are optional, unless you need more help.
You may attend any recitation(s).
Which do you think you'll attend? (you may check several)

(a) 10:00 (b) 11:00 (c) 12:00
(d) 1:00 (e) 2:00 (f) I prefer a different time (please post on piazza, but no promises)